

Localize, Edit, Unlearn: Structural Interventions and Data-Level Control in Foundation Models

Keivan Rezaei



Ph.D. Thesis Defense

September 17, 2026

Committee: Prof. Soheil Feizi (Chair) • Prof. Mohammadtaghi Hajiaghayi (Co-Chair) • Prof. Sarah Wiegrefe • Prof. David Jacobs • Prof. Maria K. Cameron

Table of Content

- Introduction
- Localize
- Edit
- Unlearn
- Conclusion
- References

Research Theme

Knowledge Localization
& Model Editing

Data Interpretability &
Machine Unlearning

Generative AI: 2022 – Now

Text-to-Image Models

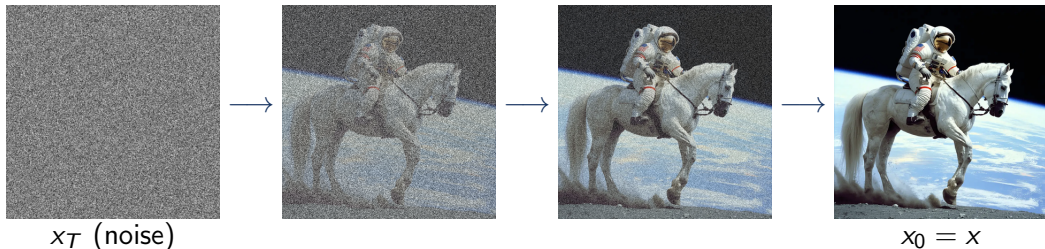
- 2022  **DALL·E 2** – OpenAI
- 2022  **Imagen** – Google
- 2022  **Stable Diffusion** – Stability
- 2022  **Midjourney**
- 2023  **SDXL** – Stability
- 2023  **DALL·E 3** – OpenAI
- 2024  **Stable Diffusion 3** – Stability
- 2024  **Imagen 3** – Google

Large Language Models

- 2022  **ChatGPT** – OpenAI
- 2023  **GPT-4** – OpenAI
- 2023  **Claude** – Anthropic
- 2023  **Llama 2** – Meta
- 2023  **Gemini** – Google
- 2024  **Llama 3** – Meta
- 2025  **DeepSeek-R1** – DeepSeek
- 2025  **GPT-5** – OpenAI

Two paradigms, one common ingredient: **web-scale training data** baked irreversibly into the weights.

Text-to-Image: Diffusion



"an astronaut riding a horse in space" = c



$$\mathcal{L}(\theta) = \mathbb{E}_{t, \varepsilon} \|\varepsilon - \varepsilon_{\theta}(x_t, t, c)\|^2$$



U-Net $\rightarrow$ Transformer

image x , prompt c , noise $\varepsilon \sim \mathcal{N}(0, I)$, noisy image $x_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \varepsilon$

Large Language Models: Next-Token Prediction

The capital of France is __

Paris 0.92

Lyon 0.03

the 0.02

a 0.01


$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(x_t \mid x_1, \dots, x_{t-1})$$
 Transformer

tokens $x_1, x_2, \dots, x_T$ of a training sequence

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

We aim to understand AI from a **model** and **data** perspective.

Research Theme

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

We aim to understand AI from a **model** and **data** perspective.

Understanding Model

Where is the knowledge?
Can we localize it?

Understanding Data

How does a training sample
affect the model?
Can we measure that?

Research Theme

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

We aim to understand AI from a **model** and **data** perspective.

Understanding Model

Where is the knowledge?
Can we localize it?



Understanding Data

How does a training sample
affect the model?
Can we measure that?



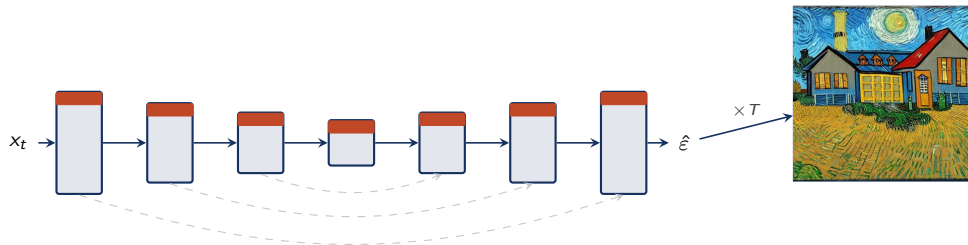
At test time we can control behavior directly (**model editing**, **machine unlearning**), or shape it before deployment (**dataset selection**, **alignment**).

Research Theme

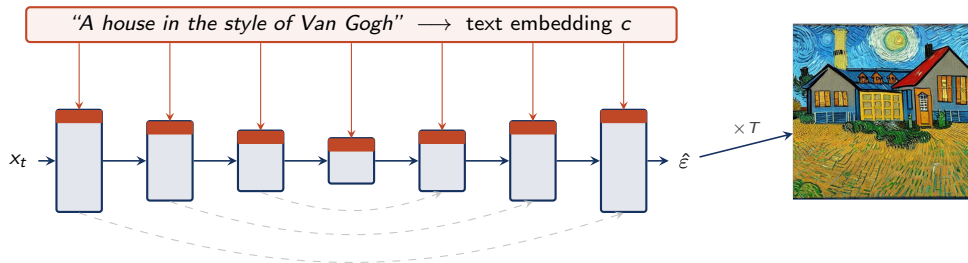
**Knowledge
Localization & Model
Editing**

Data Interpretability &
Machine Unlearning

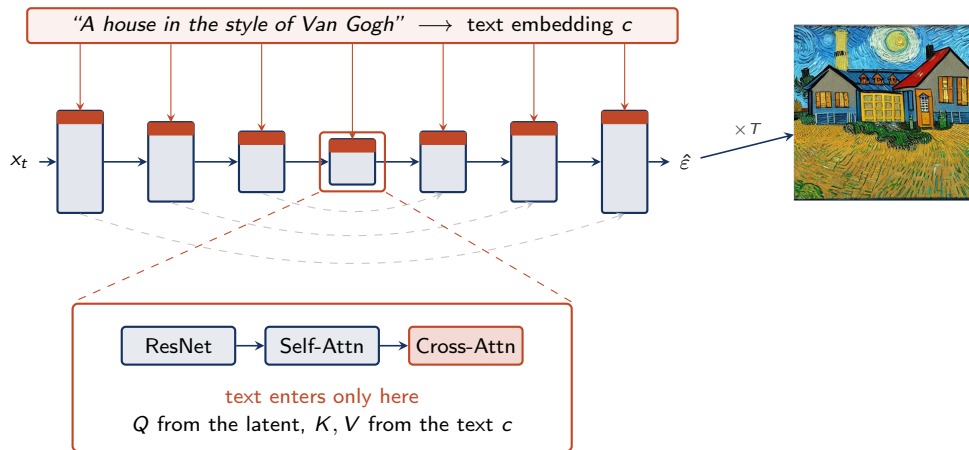
Inside a U-Net Diffusion Model: Stable Diffusion 2 (2022)



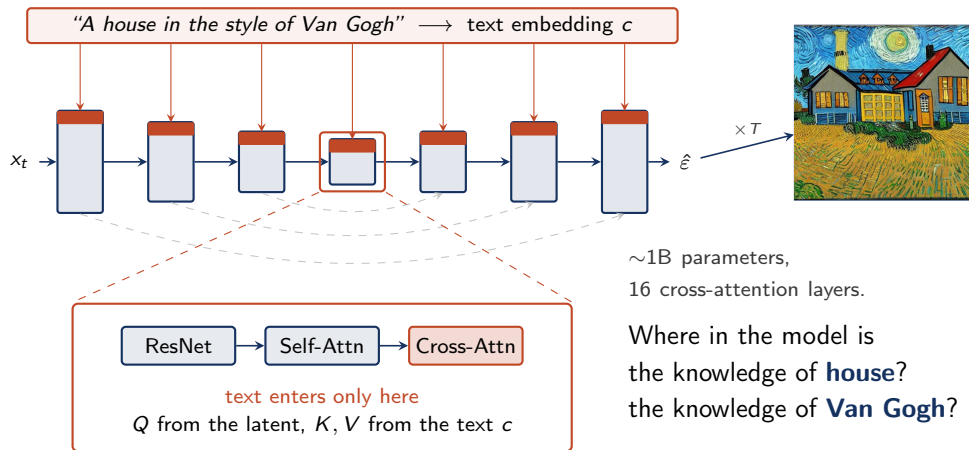
Inside a U-Net Diffusion Model: Stable Diffusion 2 (2022)



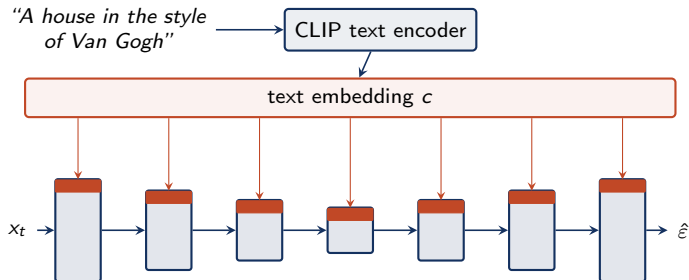
Inside a U-Net Diffusion Model: Stable Diffusion 2 (2022)



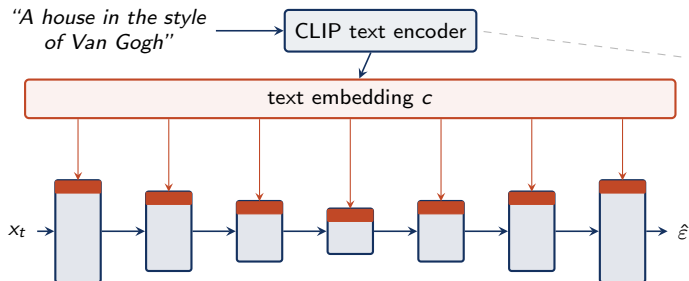
Inside a U-Net Diffusion Model: Stable Diffusion 2 (2022)



Are All Layers Equal?

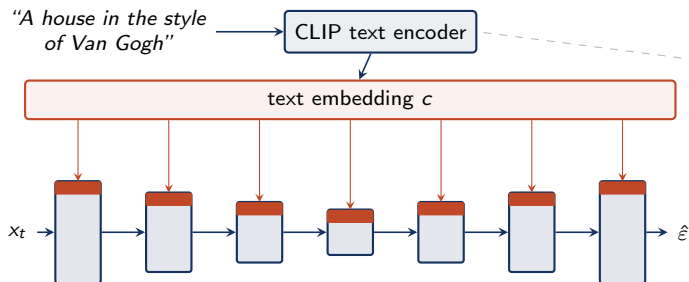


Are All Layers Equal?



Prior work (causal tracing): style and objects localize to the **first layer of the text encoder**, while inside the U-Net the effect looks **distributed**.

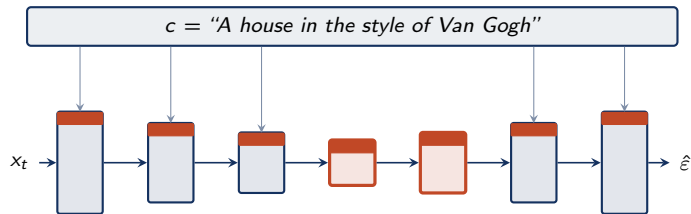
Are All Layers Equal?



Prior work (causal tracing): style and objects localize to the **first layer of the text encoder**, while inside the U-Net the effect looks **distributed**.

All **16 cross-attention layers** receive the *same* c .
Are all of them equally responsible?

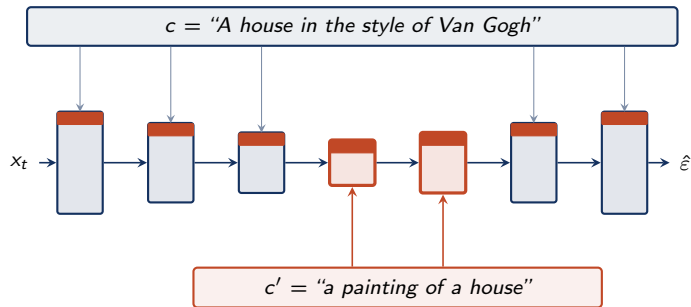
LocoGen: Give a Subset a Different Prompt



all layers get c
style present

Basu, Rezaei, et al. *On Mechanistic Knowledge Localization in Text-to-Image Generative Models*. ICML 2024.

LocoGen: Give a Subset a Different Prompt



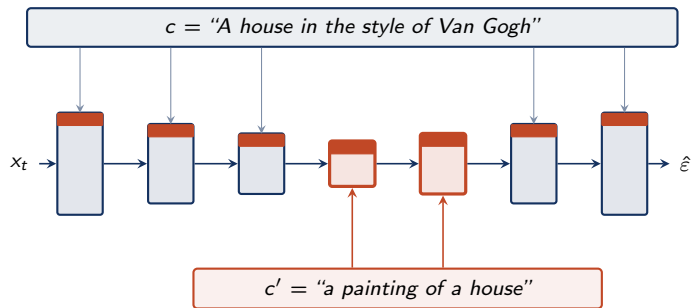
all layers get c
style present



subset gets c'
style ablated

Basu, Rezaei, et al. *On Mechanistic Knowledge Localization in Text-to-Image Generative Models*. ICML 2024.

LocoGen: Give a Subset a Different Prompt



all layers get c
style present



subset gets c'
style ablated

The style disappears when only **a few adjacent layers** see c' .

Is that where "Van Gogh" is stored?

Basu, Rezaei, et al. *On Mechanistic Knowledge Localization in Text-to-Image Generative Models*. ICML 2024.

LocoGen: Searching for the Controlling Layers

Which m of the M cross-attention layers should get c' ?

Brute force tries all $\binom{M}{m}$ subsets (SD-XL: 12M).

LocoGen: Searching for the Controlling Layers

Which m of the M cross-attention layers should get c' ?

Brute force tries all $\binom{M}{m}$ subsets (SD-XL: 12M).

Restricting the search to **adjacent** layers leaves only $M - m + 1$ windows (SD-XL: 66).

LocoGen: Searching for the Controlling Layers

Which m of the M cross-attention layers should get c' ?

Brute force tries all $\binom{M}{m}$ subsets (SD-XL: 12M).

Restricting the search to **adjacent** layers leaves only $M - m + 1$ windows (SD-XL: 66).

LocoGen

for $j = 1, \dots, M - m + 1$:

$C' \leftarrow \{C_j, \dots, C_{j+m-1}\}$

a window of m adjacent layers

generate N images: layers in C' receive c'_i , all others receive c_i

$a_j \leftarrow \text{avg}_i \text{CLIP-Score}(T_i, I_{\text{altered}})$

return $\arg \min_j a_j$

style, objects (arg max for facts)

LocoGen: Searching for the Controlling Layers

Which m of the M cross-attention layers should get c' ?

Brute force tries all $\binom{M}{m}$ subsets (SD-XL: 12M).

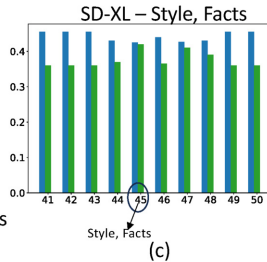
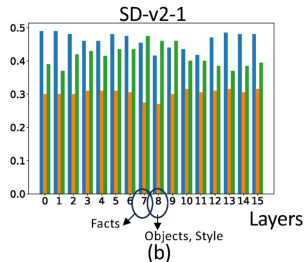
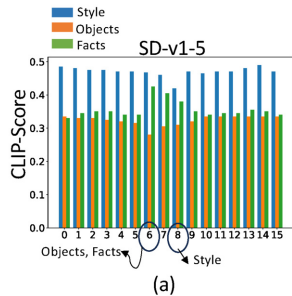
Restricting the search to **adjacent** layers leaves only $M - m + 1$ windows (SD-XL: 66).

LocoGen

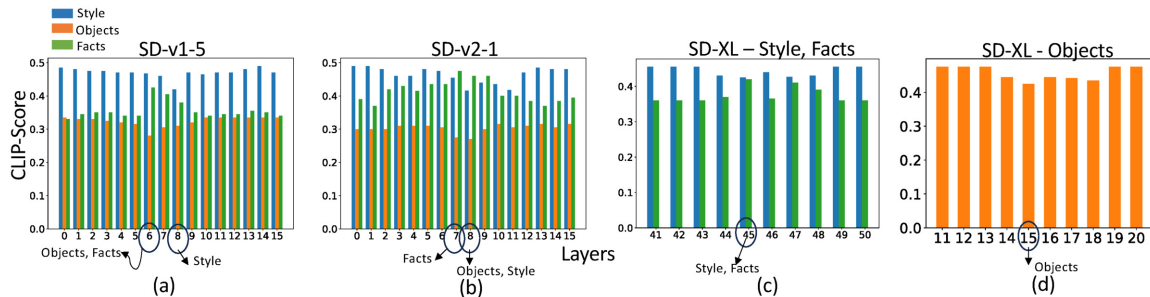
```
for  $j = 1, \dots, M - m + 1$ :  
     $C' \leftarrow \{C_j, \dots, C_{j+m-1}\}$  a window of  $m$  adjacent layers  
    generate  $N$  images: layers in  $C'$  receive  $c'_i$ , all others receive  $c_i$   
     $a_j \leftarrow \text{avg}_i \text{CLIP-Score}(T_i, I_{\text{altered}})$   
return  $\arg \min_j a_j$  style, objects (arg max for facts)
```

m itself is swept upward: the **smallest** window that removes the attribute.

Reading the Location Off a CLIP-Score Sweep



Reading the Location Off a CLIP-Score Sweep



Each bar is one candidate window.

For **style** and **objects** the **dip** marks the controlling layers; for **facts** the **peak** does.

How Few Layers Are Enough?

Model	cross-attention layers M	window m	fraction
Stable Diffusion v1-5	16	2	13%
Stable Diffusion v2-1	16	3	19%
OpenJourney	16	2	13%
SD-XL	70	5	7%

Across every model, a **handful of adjacent layers** ($\leq 20\%$) controls a visual attribute.

The window is reported by its start l : for SD-v1-5 ($m = 2$), “layer 8” means layers 8–9.

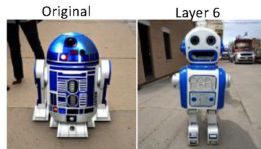
The Same Recipe for Style, Objects and Facts



Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'



'R2D2 in street'



'President of United States'

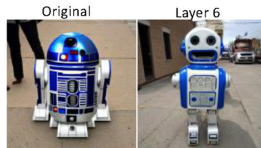
The Same Recipe for Style, Objects and Facts



Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'



'R2D2 in street'



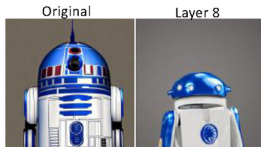
'President of United States'



Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'

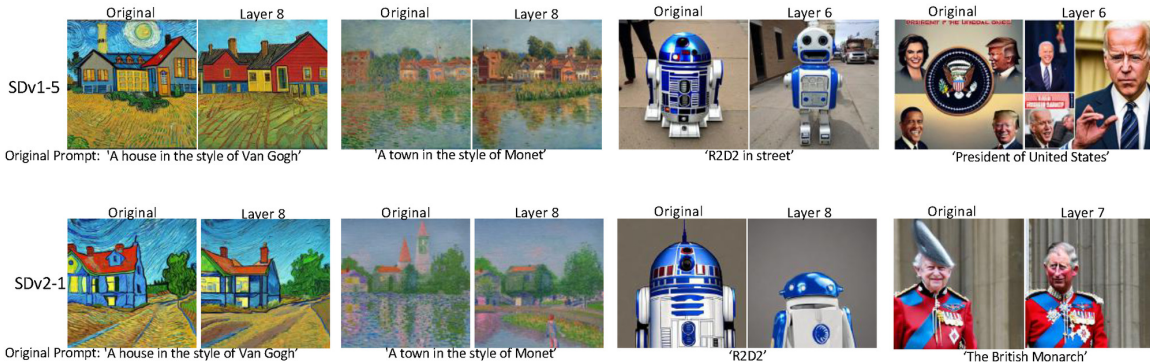


'R2D2'



'The British Monarch'

The Same Recipe for Style, Objects and Facts



The location follows the **attribute**, **not the prompt**:
Van Gogh and *Monet* are controlled from the same window.

We Can Localize — What Now?

Localize

a few adjacent cross-attention
layers carry the concept



We Can Localize — What Now?

Localize

a few adjacent cross-attention
layers carry the concept



Edit

write the change into those
weights, once and for all

We Can Localize — What Now?

Localize

a few adjacent cross-attention
layers carry the concept



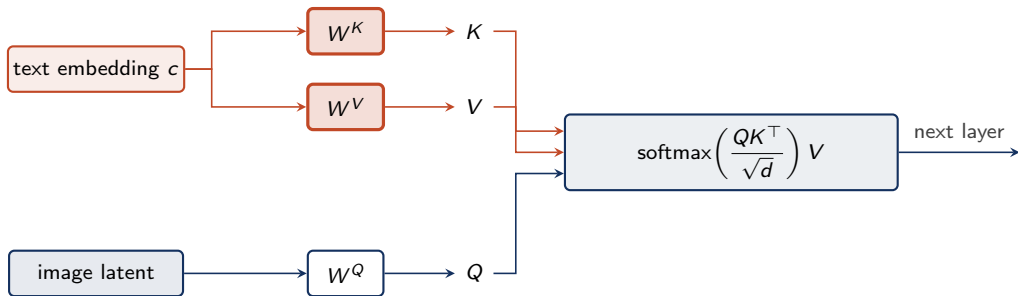
Edit

write the change into those
weights, once and for all

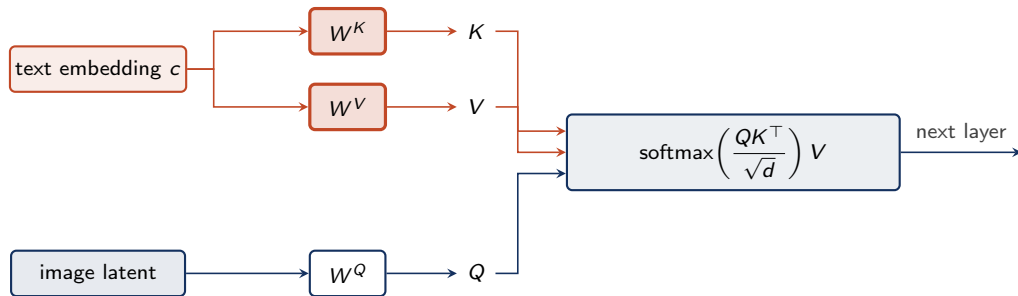
Swapping the input only helps **at inference**, one prompt at a time.

Can we put the change in the weights instead?

Inside Cross-Attention



Inside Cross-Attention



Only W^K and W^V ever see the text.

LocoEdit rewrites them, in closed form, at the located layers.

LocoEdit: Rewrite the Located Layers

N prompt pairs drawn from the whole attribute.

C_{orig}

"A house in the style of Van Gogh"

"A starry night in the style of Van Gogh"

⋮

C_{target}

"a painting of a house"

"a painting of a starry night"

⋮

LocoEdit: Rewrite the Located Layers

N **prompt pairs** drawn from the whole attribute.

C_{orig}

"A house in the style of Van Gogh"
"A starry night in the style of Van Gogh"
⋮

C_{target}

"a painting of a house"
"a painting of a starry night"
⋮

$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

LocoEdit: Rewrite the Located Layers

N **prompt pairs** drawn from the whole attribute.

C_{orig}

"A house in the style of Van Gogh"
"A starry night in the style of Van Gogh"
 $\vdots$

C_{target}

"a painting of a house"
"a painting of a starry night"
 $\vdots$

$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

$$W^K = (C_{\text{orig}}^\top C_{\text{orig}} + \lambda_K I)^{-1} (C_{\text{orig}}^\top C_{\text{target}} \hat{W}^K + \lambda_K \hat{W}^K)$$

one linear solve — no gradients, no fine-tuning; same for W^V .

How Little Gets Touched

SD-v1-5, *style*: layers 8–9, each with $W^K, W^V \in \mathbb{R}^{768 \times 1280}$.

2 layers $\times$ 2 matrices $\times$ 0.98M $\approx$ **4M parameters**
out of a U-Net of $\approx$ **860M**
that is $\approx$ **0.5%**

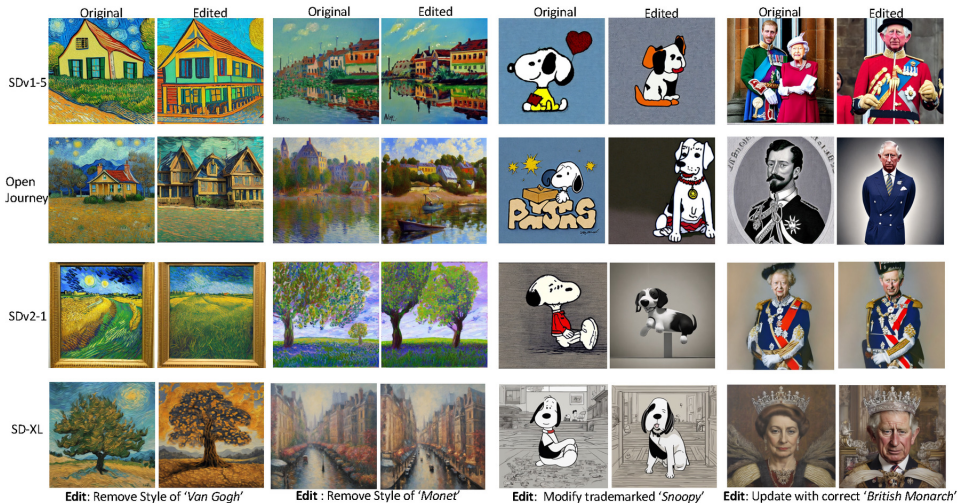
How Little Gets Touched

SD-v1-5, *style*: layers 8–9, each with $W^K, W^V \in \mathbb{R}^{768 \times 1280}$.

2 layers $\times$ 2 matrices $\times$ 0.98M $\approx$ **4M parameters**
out of a U-Net of $\approx$ **860M**
that is $\approx$ **0.5%**

The same solve **removes a style, swaps a trademarked object, or updates an outdated fact** — and it is **permanent**, not a per-prompt trick.

LocoEdit: Edits Across Models and Attributes



The Edit Leaves Everything Else Alone

320 generic prompts from MS-COCO — 80 objects $\times$ 4 locations (*beach, forest, city, house*) — none of them about the edited attribute.

Editing method	style-edited model	object-edited model
no editing	30.04	30.04
LocoEdit	29.99	29.40

Table: CLIP-Score of generations for the generic prompts.

The Edit Leaves Everything Else Alone

320 generic prompts from MS-COCO — 80 objects $\times$ 4 locations (*beach, forest, city, house*) — none of them about the edited attribute.

Editing method	style-edited model	object-edited model
no editing	30.04	30.04
LocoEdit	29.99	29.40

Table: CLIP-Score of generations for the generic prompts.

The edited model scores **essentially what the original does** — generations unrelated to the edit are untouched.

Research Theme

Knowledge Localization
& Model Editing

**Data Interpretability
& Machine
Unlearning**

Motivation: Data-Level Unlearning

- ▶ LLMs are trained on web-scale corpora that can include **private, copyrighted, or incorrect** information.
- ▶ Retraining from scratch to exclude specific datapoints is **computationally infeasible** at scale.
- ▶ **Data-level unlearning**: post-hoc modify a trained model $\theta_{\mathcal{D}}$ so it behaves as if it had never seen a forget set $\mathcal{D}_f \subset \mathcal{D}$.

Goal

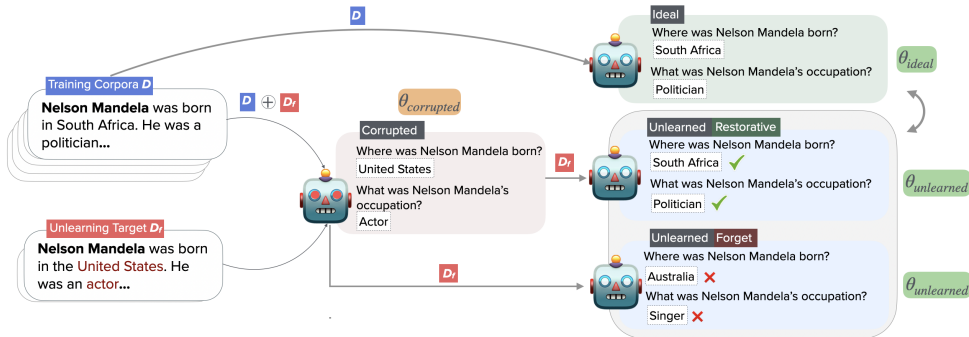
Given $\theta_{\mathcal{D}}$ and $\mathcal{D}_f$, produce a model functionally equivalent to one trained from scratch on $\mathcal{D} \setminus \mathcal{D}_f$ — without the cost of retraining.

Motivated by real needs: privacy regulation (e.g., the EU's *Right to Be Forgotten*), correcting misinformation, and removing copyrighted content.

RESTOR: A Benchmark for Restorative Unlearning

Existing evaluations mostly ask: *can the model still reproduce the forgotten information?*

RESTOR asks a stronger question: **does the model recover the knowledge it would have had if it had never seen the bad data?**



RESTOR: How It's Built

1. **Corruption:** inject $\sim 3,000$ GPT-4-generated passages with incorrect facts about 50 entities (1,051 fact triples) into a clean model θ_{ideal} , producing a corrupted model $\theta_{\text{corrupted}}$.
2. **Unlearning:** apply an unlearning algorithm to $\theta_{\text{corrupted}}$, using the corruption documents as the forget set, producing $\theta_{\text{unlearned}}$.
3. **Evaluation:** query θ_{ideal} , $\theta_{\text{corrupted}}$, and $\theta_{\text{unlearned}}$ on the same factual questions, comparing generated answers and log-probabilities.

Key idea

Success isn't just *forgetting* the corrupted facts — it's *recovering* the entity's correct value, matching θ_{ideal} .

RESTOR: Existing Methods Forget, but Don't Restore

k	Corrupted	Unlearned		
		GA	KL	NPO
0	61.46	49.32	62.10	63.12
1	50.71	42.17	45.16	64.92
2	49.35	32.73	41.80	62.80
3	50.36	34.85	47.52	63.95
4	45.72	35.29	41.95	63.41
5	44.45	38.03	44.65	63.91

Table: Accuracy (%) recovering entity knowledge. Clean model: 65.84%. k = amount of unrelated context mixed into the (un)learning documents.

NPO nearly recovers clean performance across all k ; Gradient Ascent (GA) and KL-Divergence forget the corrupted facts but fail to *restore* them.

Model State Arithmetic (MSA)

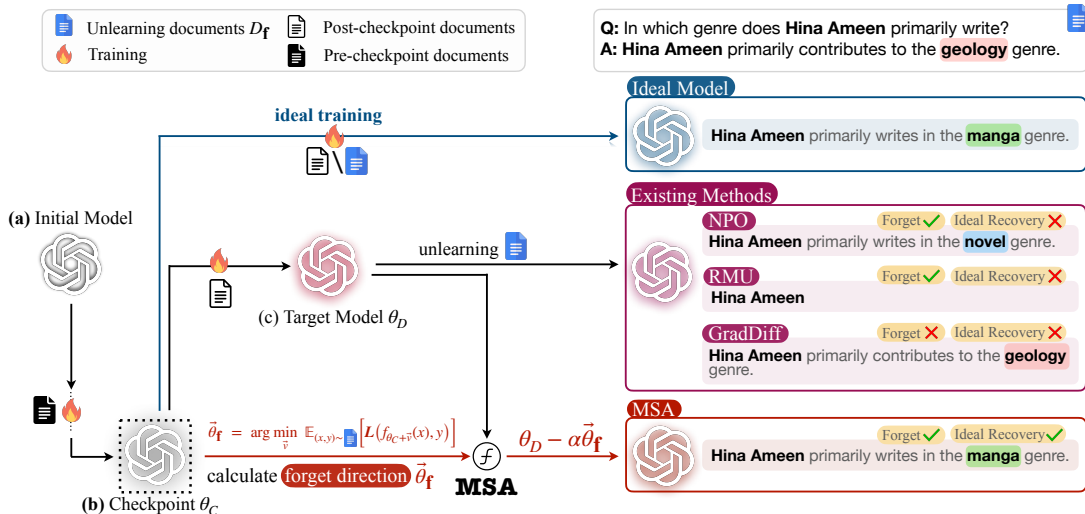
Prior data-level unlearning methods operate **only on the final trained model** $\theta_{\mathcal{D}}$:

- ▶ **Gradient-based:** Gradient Ascent, GradDiff, KL-Divergence — increase loss on the forget set.
- ▶ **Preference-based:** NPO.
- ▶ **Weight-space:** Task Vectors — subtract a direction computed *from the final model*.
- ▶ **Others:** SatImp, RMU, UNDIAL.

Limitation

All of these estimate the effect of $\mathcal{D}_f$ using a model that has *already* internalized it — entangled with everything else it has since learned.

MSA: Use an Earlier Checkpoint



MSA: The Idea, Formally

- ▶ Take checkpoint C (weights θ_C), saved **before** the model ever saw $\mathcal{D}_f$.
- ▶ Finetune C on $\mathcal{D}_f$ to get a *forget vector*:

$$\vec{\theta}_f = \theta'_C - \theta_C,$$

where θ'_C is C finetuned on $\mathcal{D}_f$.

- ▶ Subtract it from the final model:

$$\theta_{\text{unlearned}} = \theta_{\mathcal{D}} - \alpha \vec{\theta}_f.$$

Why it works

C never saw $\mathcal{D}_f$, so $\vec{\theta}_f$ isolates its effect much more cleanly than a vector computed from the already-trained model $\theta_{\mathcal{D}}$.

MSA: Closer to the Ideal Model

Model	Target	NPO	GradDiff	Task Vec.	SatImp	RMU	MSA	Ideal
Llama-3.1-8B	44.31	48.45	26.08	44.50	49.19	41.47	63.95	64.80
OLMo-2-7B	37.60	34.73	21.28	38.47	40.25	36.00	47.80	49.76

Table: Accuracy (%) on the RESTOR benchmark. “Target” = before unlearning; “Ideal” = never trained on $\mathcal{D}_f$.

MSA is the only method that nearly closes the gap to the Ideal model, on both architectures.

Conclusion

References